MIXING

Gluten is a self-organizing protein. This doesn't mean you can pour water on flour and walk away, hoping that dough will develop. You have to give it a little jump-start to distribute the ingredients evenly. That jump-start involves mixing. The most important function of mixing is to force the flour to hydrate, which unleashes a cascade of chemical reactions. Hydration, not kneading, is what allows a gluten network to develop (see page 29).

Hydration can be accomplished slowly by simply combining the ingredients and allowing the flour to absorb the water over time, or faster by manipulating the dough by hand, or in minutes or seconds by machines. The faster the mix, the faster the hydration—and the faster the dough will develop. We cover both approaches in this chapter.

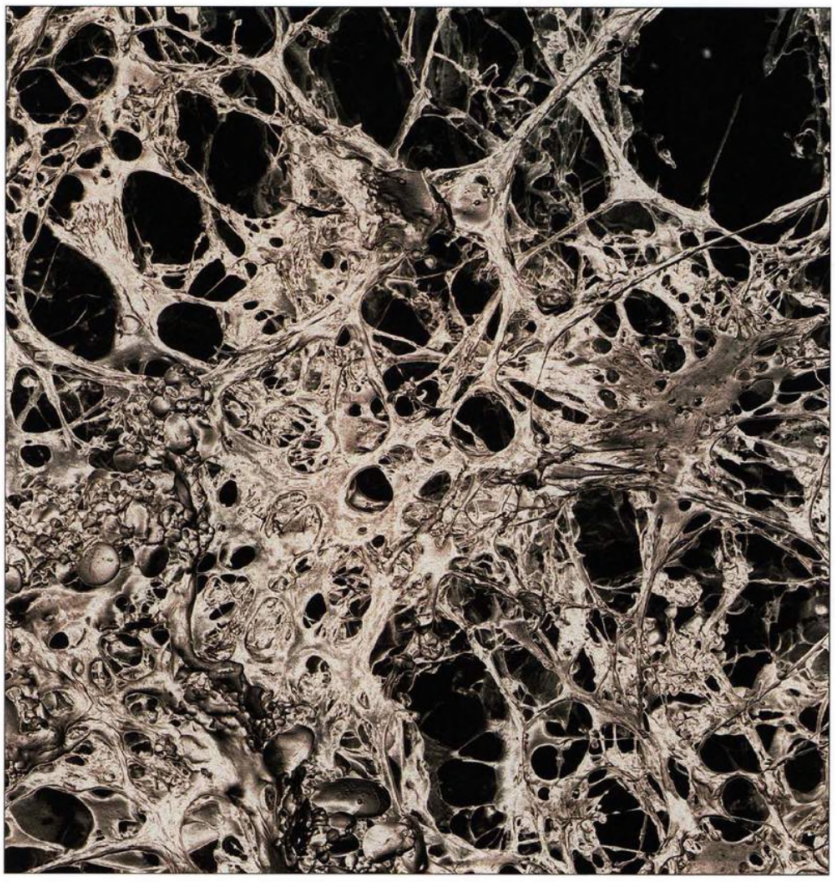
Pizza makers tend to passionately promote their own chosen mixing method above all others or insist that a specific mixer must be used for a particular type of pizza dough (such as the fork mixer for Neapolitan pizza dough). If one method works well for you, great. But the strong opinions tend to be more about personal preferences and production requirements rather than absolute rules. The reality is, there's more than one way to make very good dough, and the choice is yours. This is not to say that all mixing methods are the same (or that all mixing methods are suited for all styles of pizza dough). Machine mixing speeds up the overall process and can create a different texture in the final pizza. And it's certainly possible to overmix and create a ropy dough (see page 36).

Moreover, while hand-mixing can produce good results, it's not always practical, especially when making large amounts of dough in commercial environments. Also, most pizza doughs require mixing to full gluten development, which is daunting to do by hand. We like to let the machines do the work.

Machine mixing has a secondary effect: it creates friction, which generates heat and warms the dough. This heat helps not only with hydration but also with fermentation because yeast works faster at warmer temperatures. These factors make a mixer particularly helpful if you don't have time to let the dough fully hydrate on its own as the no-knead method requires. The amount of mixing affects how all the ingredients function together.

We mix most of our thin- and medium-crust pizza doughs to full gluten development, which means that no bulk fermentation is needed; the exception is our master Neapolitan dough, which is bulk fermented for 20–24 hours. We then divide and preshape the dough, which needs to relax for 3–4 hours before shaping and baking. The majority of our bread-like doughs are mixed to medium gluten development and then bulk fermented, with a series of folds that allow the dough to finish developing (Detroit-style pizza dough is the exception).

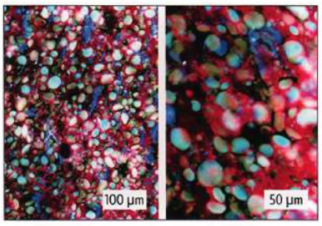
Another concept that's intertwined with mixing is autolyse (see page 32). The intent is to decrease the amount of active time spent mixing because the flour becomes hydrated in this initial rest time so the gluten development chain reaction can begin. Note that while autolyse does shorten the time spent mixing, it doesn't really save any time since the dough rests for 20–30 minutes. We found that it is not a necessary step if you are going to mix a dough to full gluten development in the mixer. However, for our high-hydration doughs that are mixed to medium gluten development, we do generally recommend an autolyse step (we do autolyse our Quad Cities pizza dough, which is mixed to full gluten development).



This image shows the gluten network at the microscopic level. We washed away the starch granules before looking at the dough under a scanning electron microscope. The resulting image shows an entangled network of gluten strands that resemble a jumble of highway interchanges.

The proofing method we developed for our master Neapolitan dough is exceptional. We discovered it only after extensive experimentation on many mixing and proofing variables (see page 88).

Although you can mix either by hand or by machine, we recommend machine mixing for all of our doughs and provide instructions for a variety of mixers in the Pizza Dough Recipes chapter on page 81.



Dough develops its structure with mixing and time. Here, dough has been stained and observed at low (left) and high (right) magnification. The gluten appears red, and the white and pale-blue discs are the starch. When dough reaches full gluten development, its components have organized into a cohesive and regular structure, with the starch granules evenly spread throughout the gluten matrix.

THE DETAILS OF MIXING

The main point of mixing is to hydrate the flour. Environmental and physical factors also play a part in the speed of gluten development and the size and distribution of bubbles in the dough. The size of your flour particles will make a difference, as will the temperature of your dough, which will depend on your choice of mixing methods (see pages 34 and 45). There are many methods, and the one you choose will affect your workflow and the characteristics of your final dough.

When flour is mixed with water, proteins in the flour called glutenin and gliadin are hydrated. They're known as gluten-forming proteins because they bind almost immediately to form gluten. Proteases (protein-snipping enzymes) cut the gluten strands into smaller pieces that are able to make additional connections (see page 29). As the gluten further develops through the mixing process, the chains become more numerous and elongated, and they organize into a cross-linked net that is both extensible and elastic. That rubbery framework will give the dough structure and allow it to expand as the yeast creates gas.

Using an electric mixer will speed up the process, which has practical importance to pizza makers. The amount of water in the mix also influences this process. The more water you add, the more extensible the dough will be, which can make it unwieldy. On the other hand, the stiffer the dough, the stronger it will be, which can make it harder to stretch. Often, the area in between those extremes is the best. The amount of water also has a direct effect on both enzymatic activity and fermentation: more water means increased activity and vice versa.

While proteases are slicing up strands of gluten, other enzymes are at work on starch. Flour typically contains only 1%–2% simple sugar, which isn't enough food for the yeasts; without greater sustenance, they can't make a dough rise because there isn't enough sugar to create carbon dioxide. The flour's starch must be broken down into simple sugars that yeasts can consume and convert. The multistep enzymatic process that breaks down the starch begins with amylase, which works on the components of damaged starch: amylose and amylopectin. (About 4% of the starch granules are broken during milling, which is an important step; an enzyme can't enter an intact starch granule.)

What happens if you don't mix? We sprinkled a few drops of water onto flour to find out. Even given an hour, the water could penetrate only a few millimeters. Then as it dried, the surface tension caused it to curl. Water stops migrating into the flour because the flour particles swell and effectively dam up the flow. Flour needs help—it's why you need to put effort into mixing.

The Mixing Trade-Off: Intensity versus Time

You can mix fast and hard or slowly and gently, but mixing is mandatory. Stir dry flour as quickly as you like; it will never turn into dough unless you add water. Or, conversely, add flour to water and let it sit in the bowl without stirring—again, the mixture will not turn into dough on its own.

The reason you have to mix flour with water to fully develop the gluten network comes down to physical chemistry. Mixing performs two indispensable operations. First, it spreads the water evenly throughout the flour, a process called dispersion. Second, mixing chemically alters the gluten proteins in the flour, a process known as hydration. To get a good, uniform dough, you have to do both operations—and dispersion has to happen either before hydration or in tandem with it.

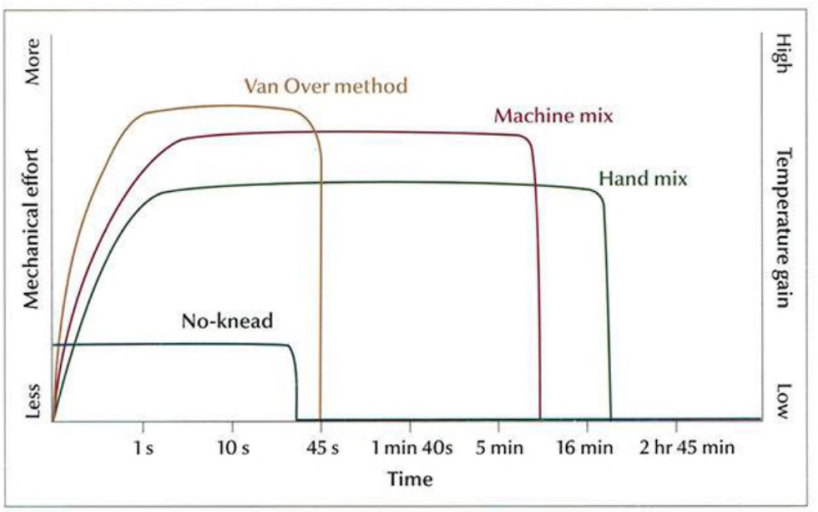
But there is no law that dictates the best mixing method, speed, or time to produce a particular kind of dough. You can get the same good results using a wide range of methods if you understand how to trade off time and intensity—and how high-energy mixing can cause side effects.

Water can move only so far on its own through a mass of flour. It needs the push of mechanical agitation to achieve full dispersion. That's why even slow-mixing techniques used for autolyse and no-knead doughs include an initial stirring step. Hydration happens quickly, but it takes time to form the chemical attachments that knit gluten proteins together into an elastic network. That's why even very high-energy mixing techniques involve some stirring.

Think of it this way: mixing = mechanical power × time. The more time you allow, the less energetic the stirring can be. You can make a no-knead dough using very little physical effort if you have hours to spare. Pros typically make the opposite trade-off: they use powerful, fast-moving mixers to shorten the timelines.

Machine mixing can have side effects that may be desirable in some situations. Friction is one of them: the faster you stir, the more the dough heats up. This can accelerate fermentation. Excessive heat also can impact fermentation. If yeasts get too warm, they feed too quickly, and flavor doesn't develop properly. Also, your dough can quickly overproof.

All mixing methods perform essentially the same operations—diffusion and hydration—but at different rates and with subtle differences in their side effects.



Mixing = mechanical power × time. This graph depicts the trade-off between mechanical power and time when mixing dough. The more powerful the mixer, the less time needed to mix the dough. However, the more powerful the mixer, the more friction is produced, which can lead to unwanted heat transfer.

Doughs mixed in a food processor (top) use a high amount of mechanical power in a very short amount of time: 45 seconds to reach full gluten development. Doughs mixed by hand (left) are on the other end of the spectrum and require a long bulk fermentation stage to allow the gluten to develop.